

Database Practicum

WEEK 6 ASSIGNMENT



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I. Student Activity

Buatlah tabel untuk sistem perpustakaan yang terdiri dari tiga tabel yaitu books, user, dan flow

dengan kolom sebagai berikut:

1. Books: bookID, bookTitle, authorName, borrowedStatus
2. Flow: flowID, userIDBorrowing, bookIDBorrowed, borrowDate, returnDate
3. User: userID, userName, numberOfBorrowing, numberOfReturning

```
MariaDB [(none)]> use library;
Database changed
MariaDB [library]> CREATE TABLE books (
->     bookID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
->     bookTitle VARCHAR(100),
->     authorName VARCHAR(100),
->     borrowedStatus BOOLEAN DEFAULT FALSE
-> );
Query OK, 0 rows affected (0.031 sec)

MariaDB [library]> CREATE TABLE user (
->     userID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
->     userName VARCHAR(100),
->     numberOfBorrowing INT,
->     numberOfReturning INT
-> );
Query OK, 0 rows affected (0.020 sec)

MariaDB [library]> CREATE TABLE flow (
->     flowID INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
->     userIDBorrowing INT,
->     bookIDBorrowed INT,
->     borrowDate DATE,
->     returnDate DATE,
->     FOREIGN KEY (userIDBorrowing) REFERENCES user(userID),
->     FOREIGN KEY (bookIDBorrowed) REFERENCES books(bookID)
-> );
Query OK, 0 rows affected (0.040 sec)
```

The tables

```

MariaDB [library]> INSERT INTO books (bookTitle, authorName, borrowedStatus) VALUES
-> ('The Great Gatsby', 'F. Scott Fitzgerald', FALSE),
-> ('To Kill a Mockingbird', 'Harper Lee', TRUE),
-> ('1984', 'George Orwell', FALSE),
-> ('Pride and Prejudice', 'Jane Austen', TRUE),
-> ('The Catcher in the Rye', 'J.D. Salinger', FALSE);
Query OK, 5 rows affected (0.008 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [library]> INSERT INTO user (userName, numberOfBorrowing, numberOfReturning) VALUES
-> ('Alice Johnson', 2, 1),
-> ('Bob Smith', 1, 1),
-> ('Charlie Brown', 0, 0),
-> ('Diana Prince', 3, 2),
-> ('Ethan Hunt', 1, 0);
Query OK, 5 rows affected (0.007 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [library]> INSERT INTO flow (userIDBorrowing, bookIDBorrowed, borrowDate, returnDate) VALUES
-> (1, 2, '2025-10-01', '2025-10-10'),
-> (2, 4, '2025-10-05', NULL),
-> (4, 1, '2025-09-20', '2025-09-30'),
-> (4, 5, '2025-10-02', NULL),
-> (5, 3, '2025-10-07', NULL);
Query OK, 5 rows affected (0.010 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

The content of the tables

Dan gunakan statement join untuk

1. Menampilkan semua judul buku dimana statusnya dipinjam (borrowed) dan tanggal meminjamannya kemarin.

```

MariaDB [library]> select b.bookTitle, b.borrowedStatus, f.borrowDate
-> from books b
-> left join flow f ON b.bookID = f.bookIDBorrowed
-> where f.borrowDate = '2025-10-09';
Empty set (0.001 sec)

```

2. Menampilkan semua judul buku bahkan buku yang tidak dipinjam dan userID peminjamnya untuk buku yang dipinjam.

```

MariaDB [library]> select b.bookTitle, b.borrowedStatus, f.userIDBorrowing
-> from books b
-> left join flow f ON b.bookID = f.bookIDBorrowed;
+-----+-----+-----+
| bookTitle | borrowedStatus | userIDBorrowing |
+-----+-----+-----+
| The Great Gatsby | 0 | 4 |
| To Kill a Mockingbird | 1 | 1 |
| 1984 | 0 | 5 |
| Pride and Prejudice | 1 | 2 |
| The Catcher in the Rye | 0 | 4 |
+-----+-----+-----+
5 rows in set (0.001 sec)

```

3. Menampilkan semua buku yang dipinjam dan semua userID apakah dia meminjam atau tidak.

```
MariaDB [library]> select b.bookTitle, b.borrowedStatus, f.userIDBorrowing
-> from books b
-> left join flow f ON b.bookID = f.bookIDBorrowed;
```

bookTitle	borrowedStatus	userIDBorrowing
The Great Gatsby	0	4
To Kill a Mockingbird	1	1
1984	0	5
Pride and Prejudice	1	2
The Catcher in the Rye	0	4

```
5 rows in set (0.001 sec)
```

4. Dengan menggunakan satu kueri, tampilkan daftar semua judul buku dan nama user yang meminjam buku tersebut dan user yang meminjam lebih dari 3 buku.

```
MariaDB [library]> SELECT b.bookTitle, u.userName
-> FROM flow f
-> JOIN books b ON f.bookIDBorrowed = b.bookID
-> JOIN user u ON f.userIDBorrowing = u.userID
-> WHERE u.numberOfBorrowing > 3;
Empty set (0.001 sec)
```

II. Exercise

1. INNER JOIN

Tampilkan daftar transaksi lengkap berisi nama pelanggan, tanggal transaksi, dan nama produk yang dibeli.

```

MariaDB [classicmodels]> select c.customerName, o.orderDate, p.productName
-> from customers c
-> inner join orders o ON c.customerNumber = o.customerNumber
-> INNER JOIN orderdetails od ON o.orderNumber = od.orderNumber
-> INNER JOIN products p ON od.productCode = p.productCode;

```

customerName	orderDate	productName
Atelier graphique	2003-05-20	1965 Aston Martin DB5
Atelier graphique	2003-05-20	1999 Indy 500 Monte Carlo SS
Atelier graphique	2003-05-20	1948 Porsche Type 356 Roadster
Atelier graphique	2003-05-20	1966 Shelby Cobra 427 S/C
Atelier graphique	2004-09-27	1996 Moto Guzzi 1100i
Atelier graphique	2004-09-27	1936 Harley Davidson El Knucklehead
Atelier graphique	2004-11-25	1938 Cadillac V-16 Presidential Limousine
Signal Gift Stores	2003-05-21	1917 Grand Touring Sedan
Signal Gift Stores	2003-05-21	1911 Ford Town Car
Signal Gift Stores	2003-05-21	1932 Model A Ford J-Coupe
Signal Gift Stores	2003-05-21	1932 Alfa Romeo 8C2300 Spider Sport
Signal Gift Stores	2003-05-21	1957 Ford Thunderbird
Signal Gift Stores	2003-05-21	1970 Chevy Chevelle SS 454
Signal Gift Stores	2003-05-21	1939 Chevrolet Deluxe Coupe
Signal Gift Stores	2003-05-21	1938 Cadillac V-16 Presidential Limousine
Signal Gift Stores	2003-05-21	1949 Jaguar XK 120
Signal Gift Stores	2003-05-21	1952 Citroen-15CV
Signal Gift Stores	2003-05-21	1969 Chevrolet Camaro Z28
Signal Gift Stores	2003-05-21	2002 Chevy Corvette
Signal Gift Stores	2003-05-21	1936 Mercedes Benz 500k Roadster
Signal Gift Stores	2004-08-06	1993 Mazda RX-7

Kelly's Gift Shop	2003-11-11	F/A 18 Hornet 1/72
Kelly's Gift Shop	2004-12-16	1980s Black Hawk Helicopter
Kelly's Gift Shop	2004-12-16	P-51-D Mustang
Kelly's Gift Shop	2004-12-16	1999 Yamaha Speed Boat
Kelly's Gift Shop	2004-12-16	1903 Ford Model A
Kelly's Gift Shop	2004-12-16	Collectable Wooden Train
Kelly's Gift Shop	2004-12-16	1941 Chevrolet Special Deluxe Cabriolet
Kelly's Gift Shop	2004-12-16	1904 Buick Runabout
Kelly's Gift Shop	2004-12-16	1928 British Royal Navy Airplane
Kelly's Gift Shop	2004-12-16	18th century schooner
Kelly's Gift Shop	2004-12-16	1900s Vintage Bi-Plane
Kelly's Gift Shop	2004-12-16	1912 Ford Model T Delivery Wagon
Kelly's Gift Shop	2004-12-16	1940 Ford Delivery Sedan
Kelly's Gift Shop	2004-12-16	The Schooner Bluenose
Kelly's Gift Shop	2004-12-16	The Mayflower
Kelly's Gift Shop	2004-12-16	The USS Constitution Ship
Kelly's Gift Shop	2004-12-16	The Titanic
Kelly's Gift Shop	2004-12-16	The Queen Mary
Kelly's Gift Shop	2004-12-16	Pont Yacht
Kelly's Gift Shop	2005-04-01	1969 Harley Davidson Ultimate Chopper
Kelly's Gift Shop	2005-04-01	1996 Moto Guzzi 1100i
Kelly's Gift Shop	2005-04-01	2003 Harley-Davidson Eagle Drag Bike
Kelly's Gift Shop	2005-04-01	2002 Suzuki XREO
Kelly's Gift Shop	2005-04-01	1936 Harley Davidson El Knucklehead
Kelly's Gift Shop	2005-04-01	1997 BMW R 1100 S
Kelly's Gift Shop	2005-04-01	1960 BSA Gold Star DBD34
Kelly's Gift Shop	2005-04-01	1997 BMW F650 ST

2996 rows in set (0.176 sec)

2. LEFT JOIN

Tampilkan semua pelanggan, termasuk pelanggan yang belum pernah melakukan Transaksi.

```
MariaDB [classicmodels]> SELECT c.customerNumber, c.customerName, o.orderNumber, o.orderDate
-> FROM customers c
-> LEFT JOIN orders o ON c.customerNumber = o.customerNumber;
```

customerNumber	customerName	orderNumber	orderDate
103	Atelier graphique	10123	2003-05-20
103	Atelier graphique	10298	2004-09-27
103	Atelier graphique	10345	2004-11-25
112	Signal Gift Stores	10124	2003-05-21
112	Signal Gift Stores	10278	2004-08-06
112	Signal Gift Stores	10346	2004-11-29
114	Australian Collectors, Co.	10120	2003-04-29
114	Australian Collectors, Co.	10125	2003-05-21
114	Australian Collectors, Co.	10223	2004-02-20
114	Australian Collectors, Co.	10342	2004-11-24
114	Australian Collectors, Co.	10347	2004-11-29
119	La Rochelle Gifts	10275	2004-07-23
119	La Rochelle Gifts	10315	2004-10-29
119	La Rochelle Gifts	10375	2005-02-03
119	La Rochelle Gifts	10425	2005-05-31
121	Baane Mini Imports	10103	2003-01-29
121	Baane Mini Imports	10158	2003-10-10
121	Baane Mini Imports	10309	2004-10-15
121	Baane Mini Imports	10325	2004-11-05
124	Mini Gifts Distributors Ltd.	10113	2003-03-26
124	Mini Gifts Distributors Ltd.	10135	2003-07-02

465	Anton Designs, Ltd.	NULL	NULL
471	Australian Collectables, Ltd	10193	2003-11-21
471	Australian Collectables, Ltd	10265	2004-07-02
471	Australian Collectables, Ltd	10415	2005-05-09
473	Frau da Collezione	10157	2003-10-09
473	Frau da Collezione	10218	2004-02-09
475	West Coast Collectables Co.	10199	2003-12-01
475	West Coast Collectables Co.	10215	2004-01-29
477	Mit Vergnügen & Co.	NULL	NULL
480	Kremlin Collectables, Co.	NULL	NULL
481	Raanan Stores, Inc	NULL	NULL
484	Iberia Gift Imports, Corp.	10184	2003-11-14
484	Iberia Gift Imports, Corp.	10303	2004-10-06
486	Motor Mint Distributors Inc.	10109	2003-03-10
486	Motor Mint Distributors Inc.	10236	2004-04-03
486	Motor Mint Distributors Inc.	10331	2004-11-17
487	Signal Collectibles Ltd.	10149	2003-09-12
487	Signal Collectibles Ltd.	10219	2004-02-10
489	Double Decker Gift Stores, Ltd	10186	2003-11-14
489	Double Decker Gift Stores, Ltd	10213	2004-01-22
495	Diecast Collectables	10207	2003-12-09
495	Diecast Collectables	10243	2004-04-26
496	Kelly's Gift Shop	10138	2003-07-07
496	Kelly's Gift Shop	10179	2003-11-11
496	Kelly's Gift Shop	10360	2004-12-16
496	Kelly's Gift Shop	10399	2005-04-01

350 rows in set (0.001 sec)

3. RIGHT JOIN

Tampilkan semua produk, termasuk produk yang belum pernah terjual.

```

MariaDB [classicmodels]> select od.orderNumber, p.productName
-> from orderdetails od
-> RIGHT JOIN products p ON od.productCode = p.productCode;

```

orderNumber	productName
10107	1969 Harley Davidson Ultimate Chopper
10121	1969 Harley Davidson Ultimate Chopper
10134	1969 Harley Davidson Ultimate Chopper
10145	1969 Harley Davidson Ultimate Chopper
10159	1969 Harley Davidson Ultimate Chopper
10168	1969 Harley Davidson Ultimate Chopper
10180	1969 Harley Davidson Ultimate Chopper
10188	1969 Harley Davidson Ultimate Chopper
10201	1969 Harley Davidson Ultimate Chopper
10211	1969 Harley Davidson Ultimate Chopper
10223	1969 Harley Davidson Ultimate Chopper
10237	1969 Harley Davidson Ultimate Chopper
10251	1969 Harley Davidson Ultimate Chopper
10263	1969 Harley Davidson Ultimate Chopper
10275	1969 Harley Davidson Ultimate Chopper
10285	1969 Harley Davidson Ultimate Chopper
10299	1969 Harley Davidson Ultimate Chopper
10309	1969 Harley Davidson Ultimate Chopper
10318	1969 Harley Davidson Ultimate Chopper
10329	1969 Harley Davidson Ultimate Chopper
10341	1969 Harley Davidson Ultimate Chopper
10354	1969 Harley Davidson Ultimate Chopper
10361	1969 Harley Davidson Ultimate Chopper

```

10117 | Pont Yacht
10129 | Pont Yacht
10142 | Pont Yacht
10153 | Pont Yacht
10167 | Pont Yacht
10177 | Pont Yacht
10185 | Pont Yacht
10197 | Pont Yacht
10208 | Pont Yacht
10222 | Pont Yacht
10232 | Pont Yacht
10248 | Pont Yacht
10261 | Pont Yacht
10273 | Pont Yacht
10283 | Pont Yacht
10293 | Pont Yacht
10306 | Pont Yacht
10315 | Pont Yacht
10327 | Pont Yacht
10337 | Pont Yacht
10350 | Pont Yacht
10360 | Pont Yacht
10373 | Pont Yacht
10386 | Pont Yacht
10397 | Pont Yacht
10414 | Pont Yacht
+-----+-----+
2997 rows in set (0.011 sec)

```

4. SELF JOIN

Dari tabel Produk, tampilkan pasangan produk yang harganya sama (kecuali dirinya sendiri).

```

MariaDB [classicmodels]> SELECT
->   p1.productCode AS Kode_Produk_1,
->   p1.productName AS Nama_Produk_1,
->   p2.productCode AS Kode_Produk_2,
->   p2.productName AS Nama_Produk_2,
->   p1.buyPrice AS Harga_Sama
-> FROM products p1, products p2
-> WHERE p1.buyPrice = p2.buyPrice AND p1.productCode <> p2.productCode;
+-----+-----+-----+-----+-----+
| Kode_Produk_1 | Nama_Produk_1 | Kode_Produk_2 | Nama_Produk_2 | Harga_Sama |
+-----+-----+-----+-----+-----+
| S18_3232      | 1992 Ferrari 360 Spider red | S12_1666      | 1958 Setra Bus | 77.90      |
| S72_3212      | Pont Yacht    | S18_2248      | 1911 Ford Town Car | 33.30      |
| S12_1666      | 1958 Setra Bus | S18_3232      | 1992 Ferrari 360 Spider red | 77.90      |
| S18_2248      | 1911 Ford Town Car | S72_3212      | Pont Yacht    | 33.30      |
+-----+-----+-----+-----+-----+
4 rows in set (0.006 sec)

```

5. UNION

Gabungkan daftar semua nama pelanggan dan semua nama supplier ke dalam satu hasil query.


```
MariaDB [classicmodels]> select customerName from customers
-> union
-> select productVendor from products;
```

customerName
Atelier graphique
Signal Gift Stores
Australian Collectors, Co.
La Rochelle Gifts
Baane Mini Imports
Mini Gifts Distributors Ltd.
Havel & Zbyszek Co
Blauer See Auto, Co.
Mini Wheels Co.
Land of Toys Inc.
Euro+ Shopping Channel
Volvo Model Replicas, Co
Danish Wholesale Imports
Saveley & Henriot, Co.
Dragon Souvenirs, Ltd.
Muscle Machine Inc
Diecast Classics Inc.
Technics Stores Inc.
Handji Gifts& Co
Herkku Gifts
American Souvenirs Inc
Porto Imports Co.
Daedalus Designs Imports

Anton Designs, Ltd.	
Australian Collectables, Ltd	
Frau da Collezione	
West Coast Collectables Co.	
Mit Vergnügen & Co.	
Kremlin Collectables, Co.	
Raanan Stores, Inc	
Iberia Gift Imports, Corp.	
Motor Mint Distributors Inc.	
Signal Collectibles Ltd.	
Double Decker Gift Stores, Ltd	
Diecast Collectables	
Kelly's Gift Shop	
Min Lin Diecast	
Classic Metal Creations	
Highway 66 Mini Classics	
Red Start Diecast	
Motor City Art Classics	
Second Gear Diecast	
Autoart Studio Design	
Welly Diecast Productions	
Unimax Art Galleries	
Studio M Art Models	
Exoto Designs	
Gearbox Collectibles	
Carousel DieCast Legends	

+-----+
135 rows in set (0.002 sec)